Week 5 cheat sheet: the ML core, closed with the committee memo

The week's picture, complete: frame, split, features, compare, justify; monitoring comes in Week 14



Friday's hard terms

Term	Plain meaning	Rule or formula	Trap, or what flips it
Model justification	One page a committee can approve	Claim, evidence, caveat, action; each with a number	Opening with AUC
Error bill	What being wrong costs	`wasted x Rs 40 + missed x Rs 1,200`	Counting found buyers as a cost
Cost cut	The cheapest threshold	`cheap / (cheap + expensive)` : 0.032	Flips when scores are stale or a budget binds
Budget cut	Act on the top k only	Compare buyers in the top k	Mixing it up with the cost cut
Paired bootstrap	Is a gap bigger than luck?	Resample, recount, read the band	A tie flips once the band clears zero
Calibration	Does 16 in 100 mean 16 in 100?	Promise against reality	High AUC is not calibration
Staleness rule	When to stop trusting the model	Number, ceiling, window, owner	"Retrain every month"
Luck band	How far a share moves by chance	`2 x sqrt(p x (1 - p) / n)`	One month over is not drift
Weakest segment	Where the model ranks worst	Weakest group with a readable band	The lowest number on a tiny group
Transfer paragraph	What carries over to a new business	Changes, stays, would leak	"Lending needs a better model"
The envelope	The sealed test slice	Opened once, after choosing	A second look makes it validation

Everything so far, one line each

Week	Day	The one line
1	Mon, Tue	The revenue tree first; like with like: same window, same denominator
1	Wed, Thu	Profile before you analyse; real or noise, sample size, the four-part note
2	Mon to Wed	SQL: selection to CTEs, joins without fan-out, window functions
2	Thu, Fri	pandas groupby, merge and reshape; Excel for the leadership deck
3	Build 1	Profile, clean, reconcile, and one headline claim defended
4	Mon, Tue	A metric that cannot be gamed; one slide for three people who disagree
4	Wed, Thu	Lift beyond chance; cohorts and funnels find where the loss happens
4	Fri	A forecast must beat a baseline that is hard to beat
5	Mon	Is it an ML problem; the split, the seal, the 88.7 percent baseline
5	Tue	Four boxes, precision and recall, the cost cut and the budget cut
5	Wed	Known at prediction time, or a leak: `duration` lifts AUC 0.80 to 0.94
5	Thu	Memorise or learn; cross-validate; the simpler model wins a tie
5	Fri	The memo: claim, evidence, caveat, action, and the transfer to lending

The week's interview anchors

Tag	Question	The shape of the answer
[S]	Walk me through a model you built and why you chose it	Claim, evidence against a baseline, caveat, action, one choice of yours
[F]	How would you know your model has gone stale?	Number, ceiling, window, owner
[F]	What would change if this moved to a lending business?	Target, costs, cut change; split and bill stay; post-decision columns leak
[D]	The committee wants the higher score; you recommend the simpler one	The gap in buyers and rupees, its band, the flip

Three facts that make it stick

- Zillow's forecast had to hold three to six months out; its 8-K of 2 November 2021 recorded a $304.4 million write-down of Zillow Offers inventory: https://www.sec.gov/Archives/edgar/data/1617640/000161764021000085/z-20211102.htm (verified 28 October 2026)
- In March 2020 the accuracy of Instacart's found-items metric dropped from 93 to 61 percent, the signal to retrain: https://fortune.com/2020/06/09/instacart-coronavirus-artificial-intelligence (verified 28 October 2026)
- The bank file's own notes say `duration` "should be discarded if the intention is to have a realistic predictive model": https://archive.ics.uci.edu/dataset/222/bank+marketing (verified 28 October 2026)